

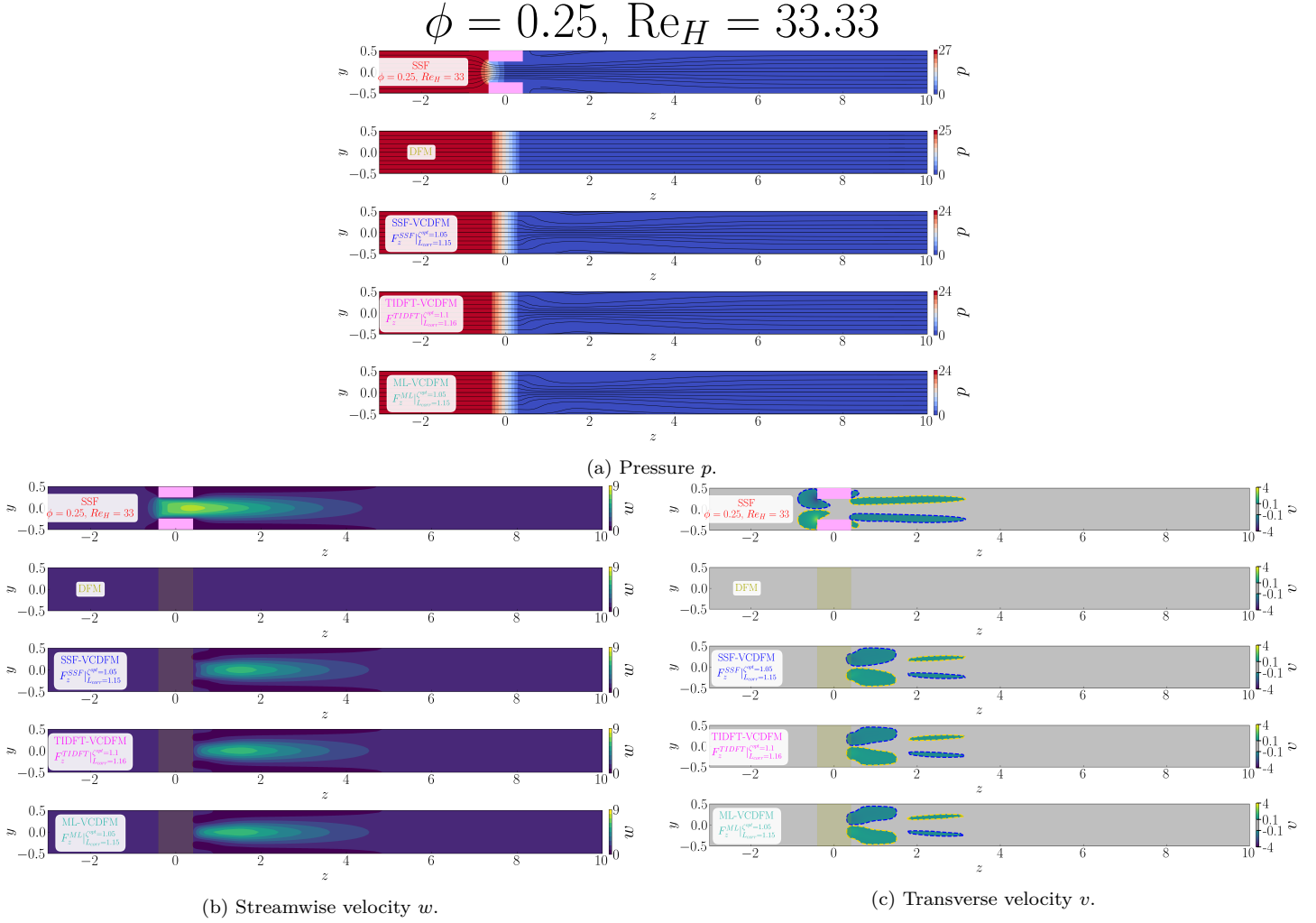


Figure D.202: Flow organization in the streamwise (z, y) -plane for the 1×1 SRG at $\phi = 0.25$, $\text{Re}_H = 33.33$, and $\alpha = 0^\circ$. The blocks show (a) pressure p with streamlines, (b) streamwise velocity w , and (c) transverse velocity v . Within each block, the rows show, from top to bottom, the SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM. Pink shading marks the grid walls in the SSF; olive shading marks the equivalent porous zone in the model solutions.

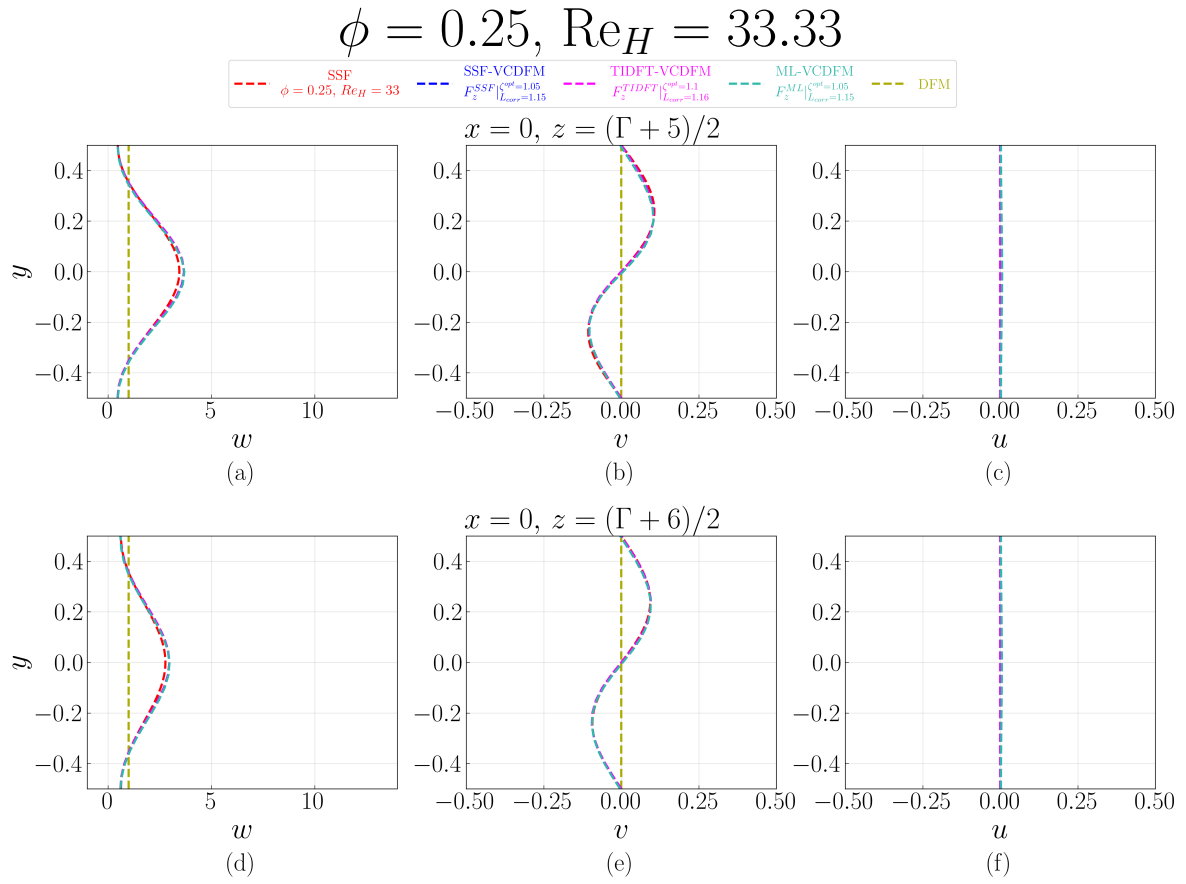


Figure D.203: Velocity profiles at $x = 0$ for the 1×1 SRG at $\phi = 0.25$ and $\text{Re}_H = 33.33$, comparing the SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM. Profiles of $w(y)$, $v(y)$, and $u(y)$ are shown at $z = (\Gamma + 5)/2$ in (a–c) and $z = (\Gamma + 6)/2$ in (d–f).

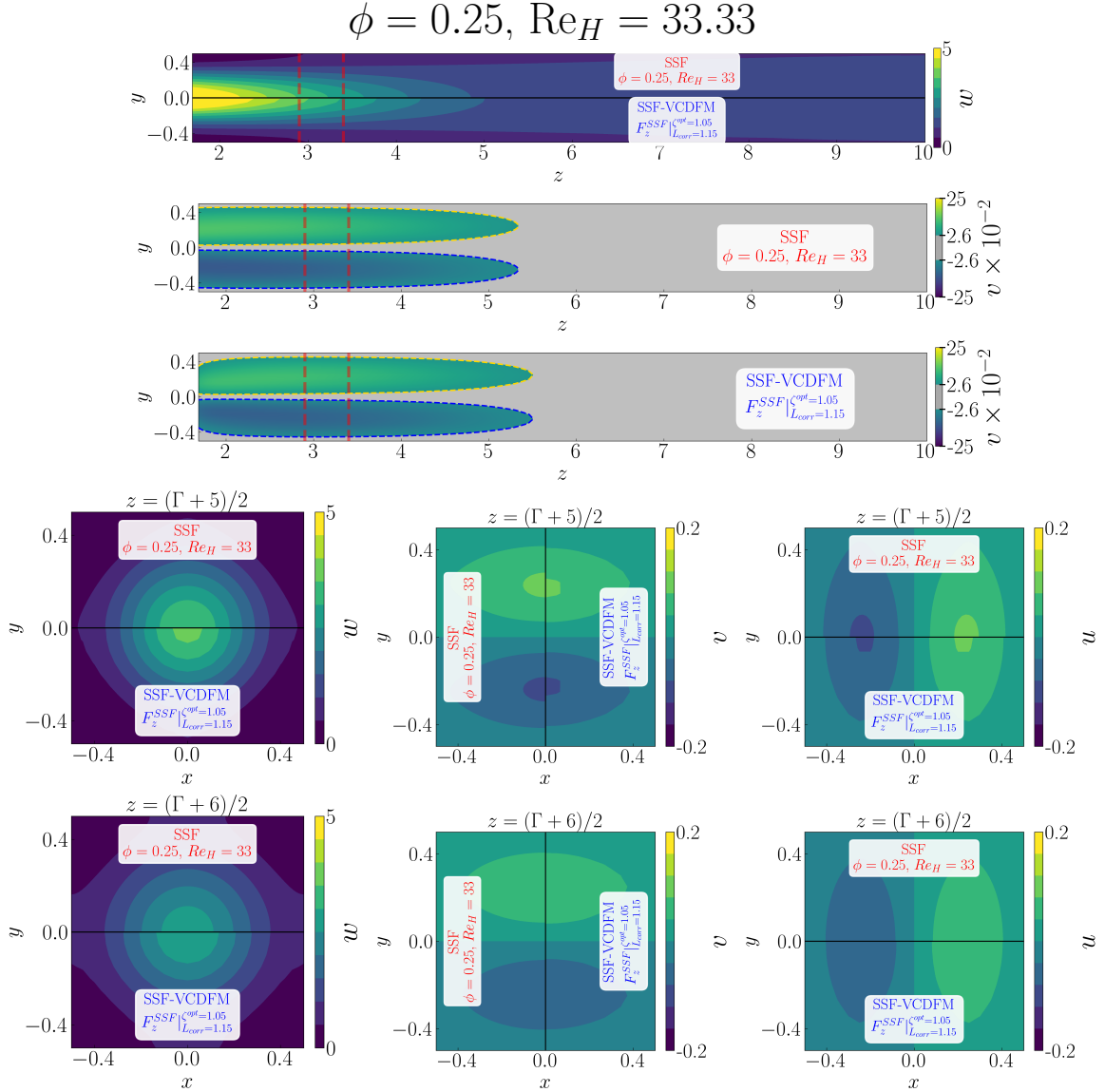


Figure D.204: Detailed comparison between the geometry-resolved small-scale flow (SSF) and SSF-VCDFM using F_z^{SSF} for the 1×1 SRG at $\phi = 0.25$ and $\text{Re}_H = 33.33$. The upper panels compare the streamwise velocity w and transverse velocity v in the central (z, y) -plane, restricted to the correction region. The lower panels compare the corresponding transverse (x, y) -plane distributions of w , v , and u , from left to right, at $z = (\Gamma + 5)/2$ in the first row and $z = (\Gamma + 6)/2$ in the second row.

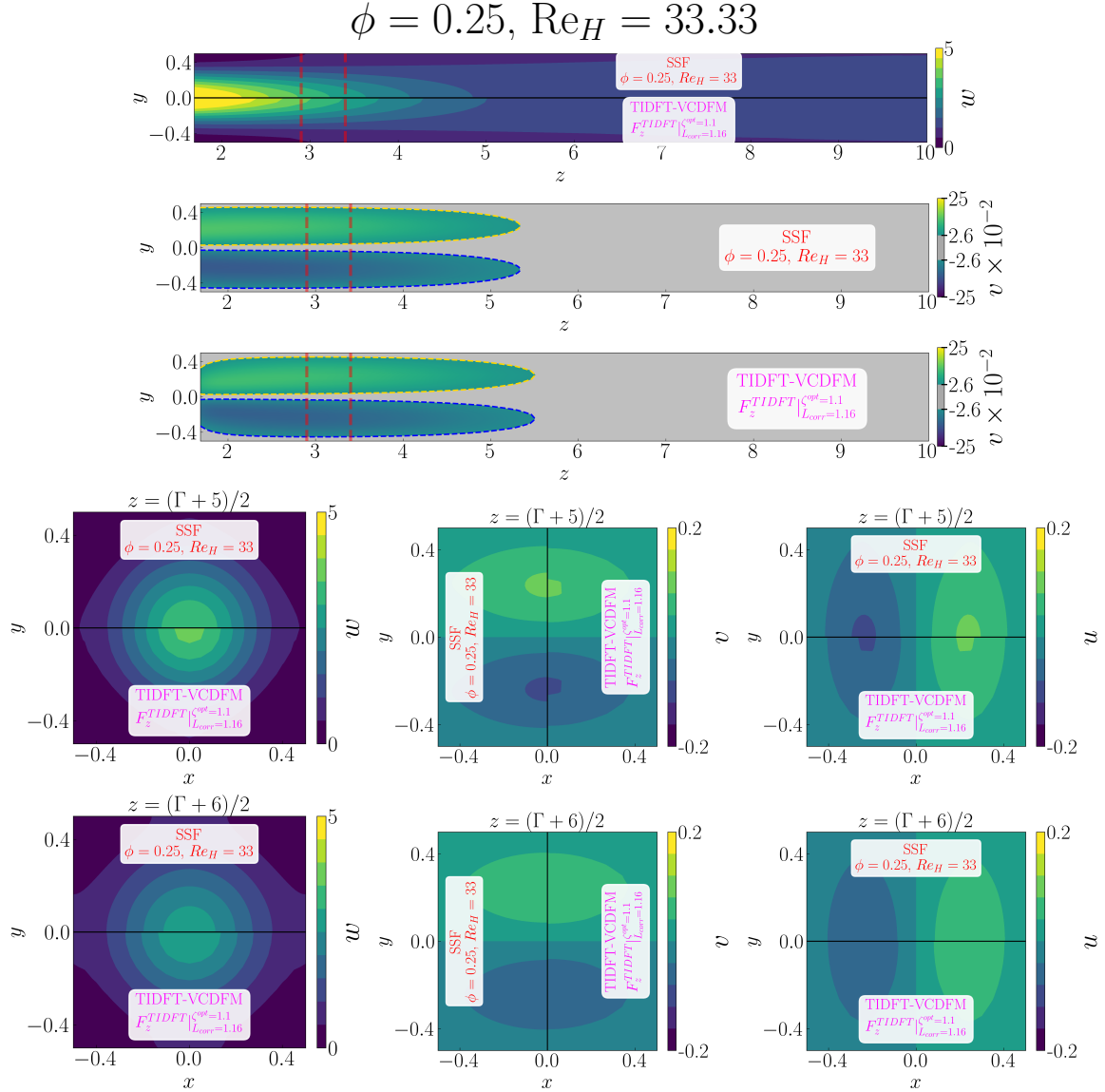


Figure D.205: Detailed comparison between the geometry-resolved small-scale flow (SSF) and TIDFT-VCDFM using F_z^{TIDFT} for the 1×1 SRG at $\phi = 0.25$ and $\text{Re}_H = 33.33$. The upper panels compare the streamwise velocity w and transverse velocity v in the central (z, y) -plane, restricted to the correction region. The lower panels compare the corresponding transverse (x, y) -plane distributions of w , v , and u , from left to right, at $z = (\Gamma + 5)/2$ in the first row and $z = (\Gamma + 6)/2$ in the second row.

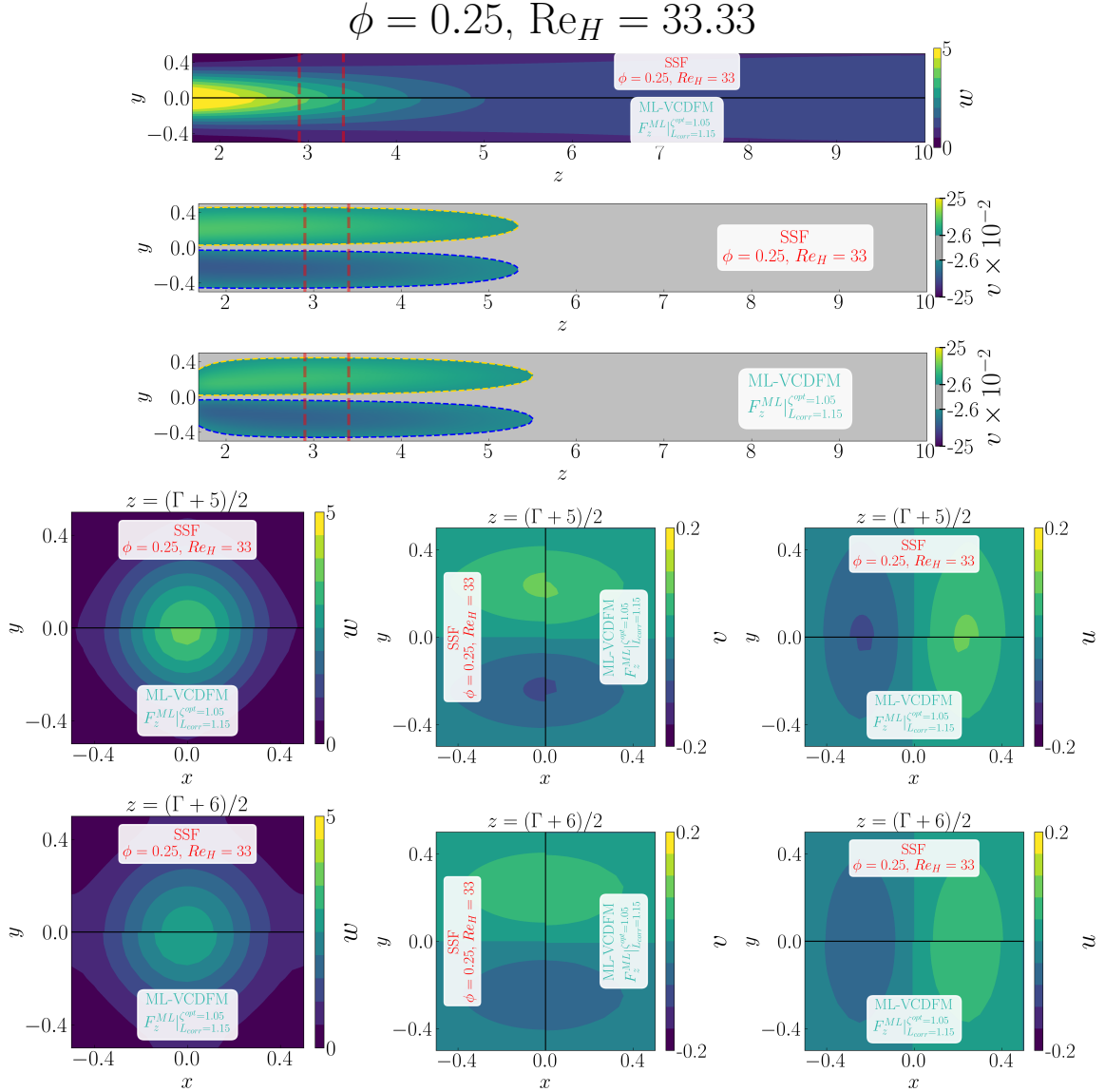


Figure D.206: Detailed comparison between the geometry-resolved small-scale flow (SSF) and ML-VCDFM using F_z^{ML} for the 1×1 SRG at $\phi = 0.25$ and $\text{Re}_H = 33.33$. The upper panels compare the streamwise velocity w and transverse velocity v in the central (z, y) -plane, restricted to the correction region. The lower panels compare the corresponding transverse (x, y) -plane distributions of w , v , and u , from left to right, at $z = (\Gamma + 5)/2$ in the first row and $z = (\Gamma + 6)/2$ in the second row.

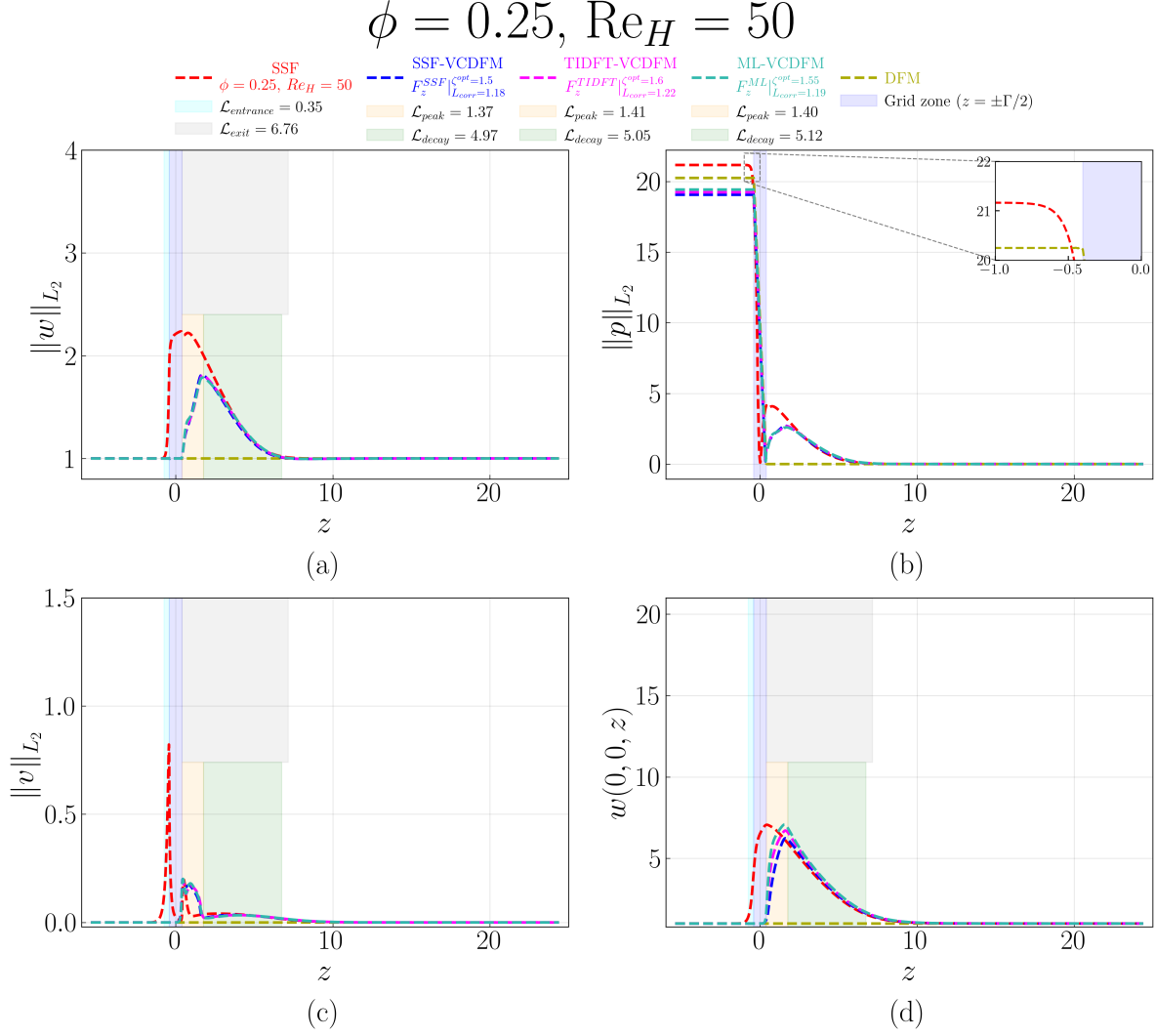


Figure D.207: Streamwise evolution of (a) $\|w\|_{L_2}$, (b) $\|p\|_{L_2}$, (c) $\|v\|_{L_2}$, and (d) centerline velocity $w(0,0,z)$ for the 1×1 SRG at $\phi = 0.25$ and $\text{Re}_H = 50$. The figure compares the SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM. Shading marks the grid and the characteristic entrance, exit, peak, and decay regions. Only the SSF-based $\mathcal{L}_{\text{peak}}$ and $\mathcal{L}_{\text{decay}}$ are marked on the plot; the corresponding TIDFT-VCDFM and ML-VCDFM values appear in the legend.

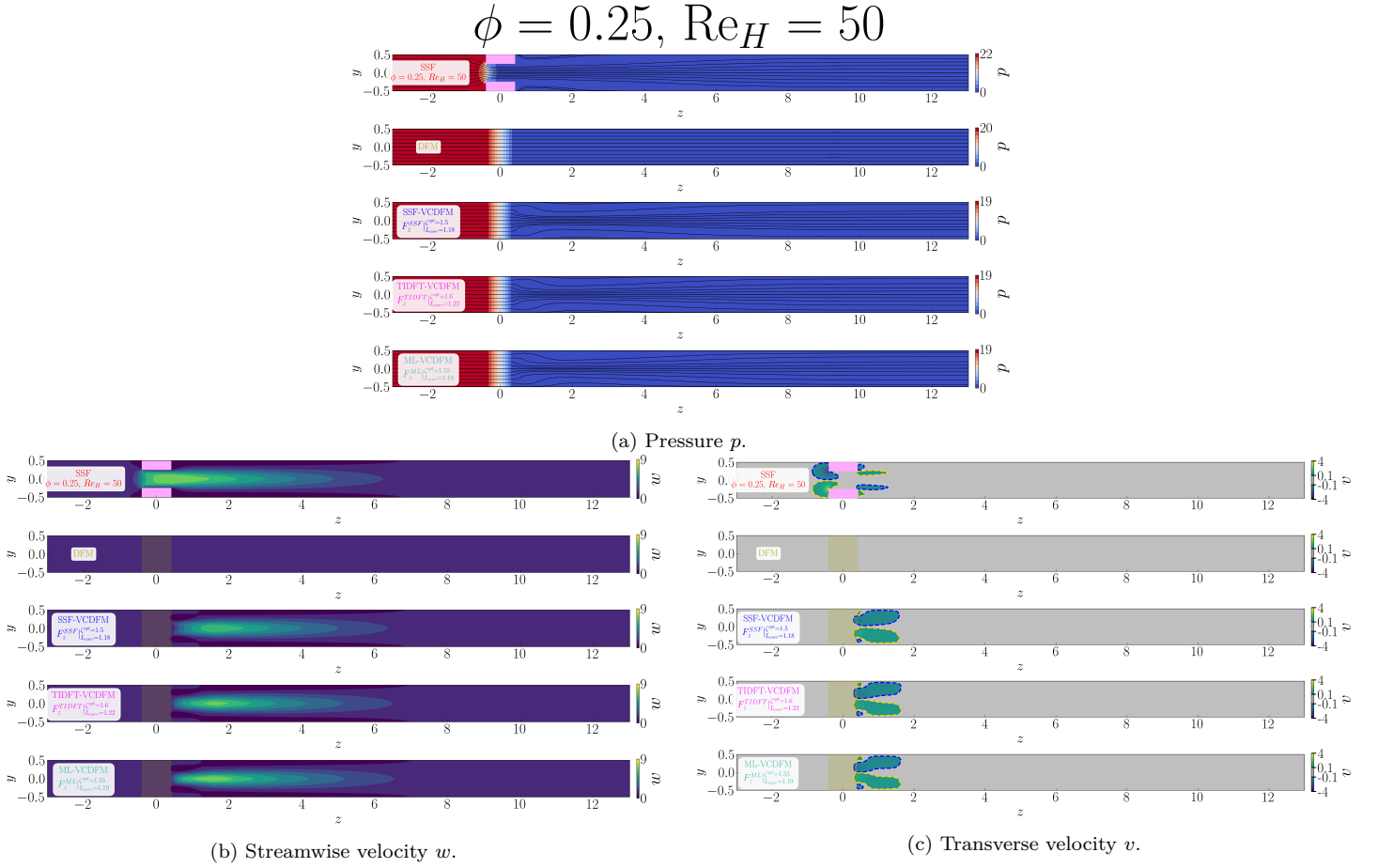


Figure D.208: Flow organization in the streamwise (z, y) -plane for the 1×1 SRG at $\phi = 0.25$, $\text{Re}_H = 50$, and $\alpha = 0^\circ$. The blocks show (a) pressure p with streamlines, (b) streamwise velocity w , and (c) transverse velocity v . Within each block, the rows show, from top to bottom, the SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM. Pink shading marks the grid walls in the SSF; olive shading marks the equivalent porous zone in the model solutions.

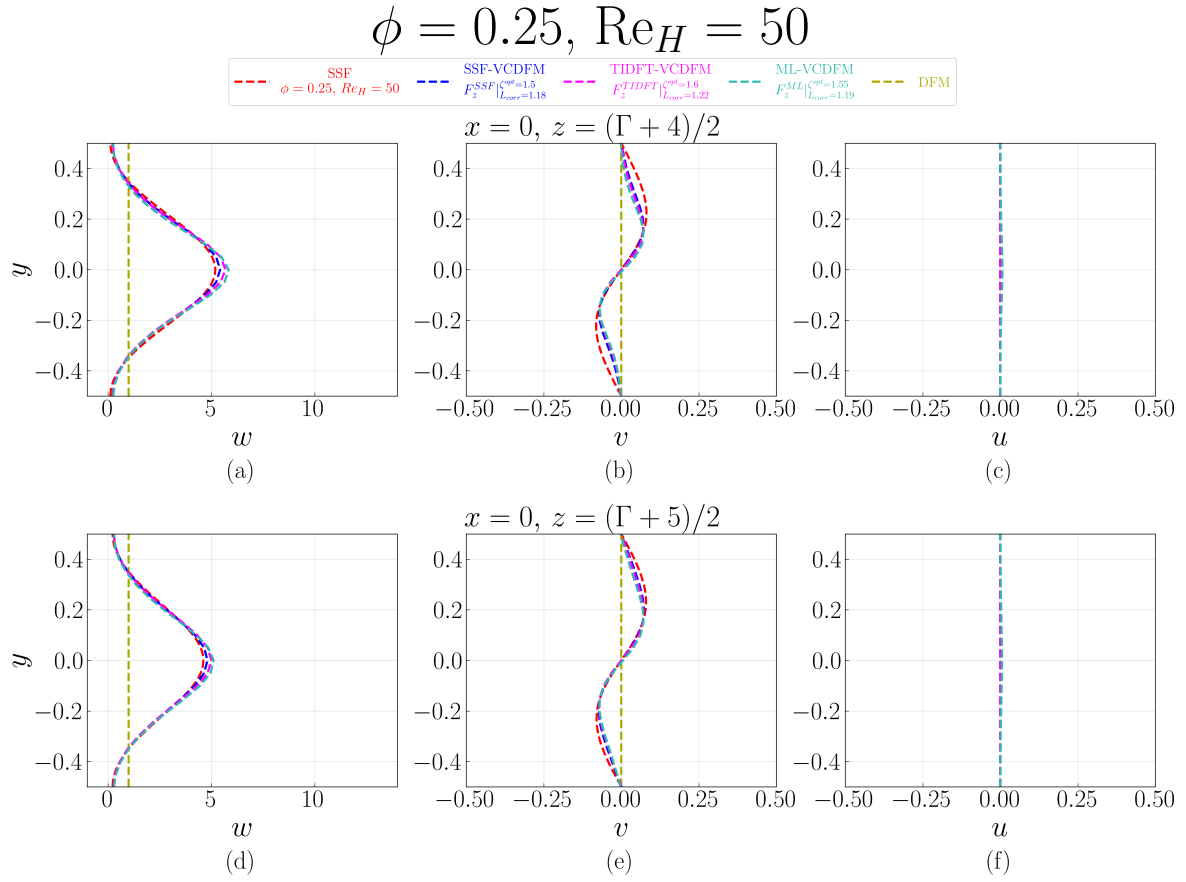


Figure D.209: Velocity profiles at $x = 0$ for the 1×1 SRG at $\phi = 0.25$ and $\text{Re}_H = 50$, comparing the SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM. Profiles of $w(y)$, $v(y)$, and $u(y)$ are shown at $z = (2\Gamma + 9)/4$ in (a–c) and $z = (2\Gamma + 11)/4$ in (d–f).

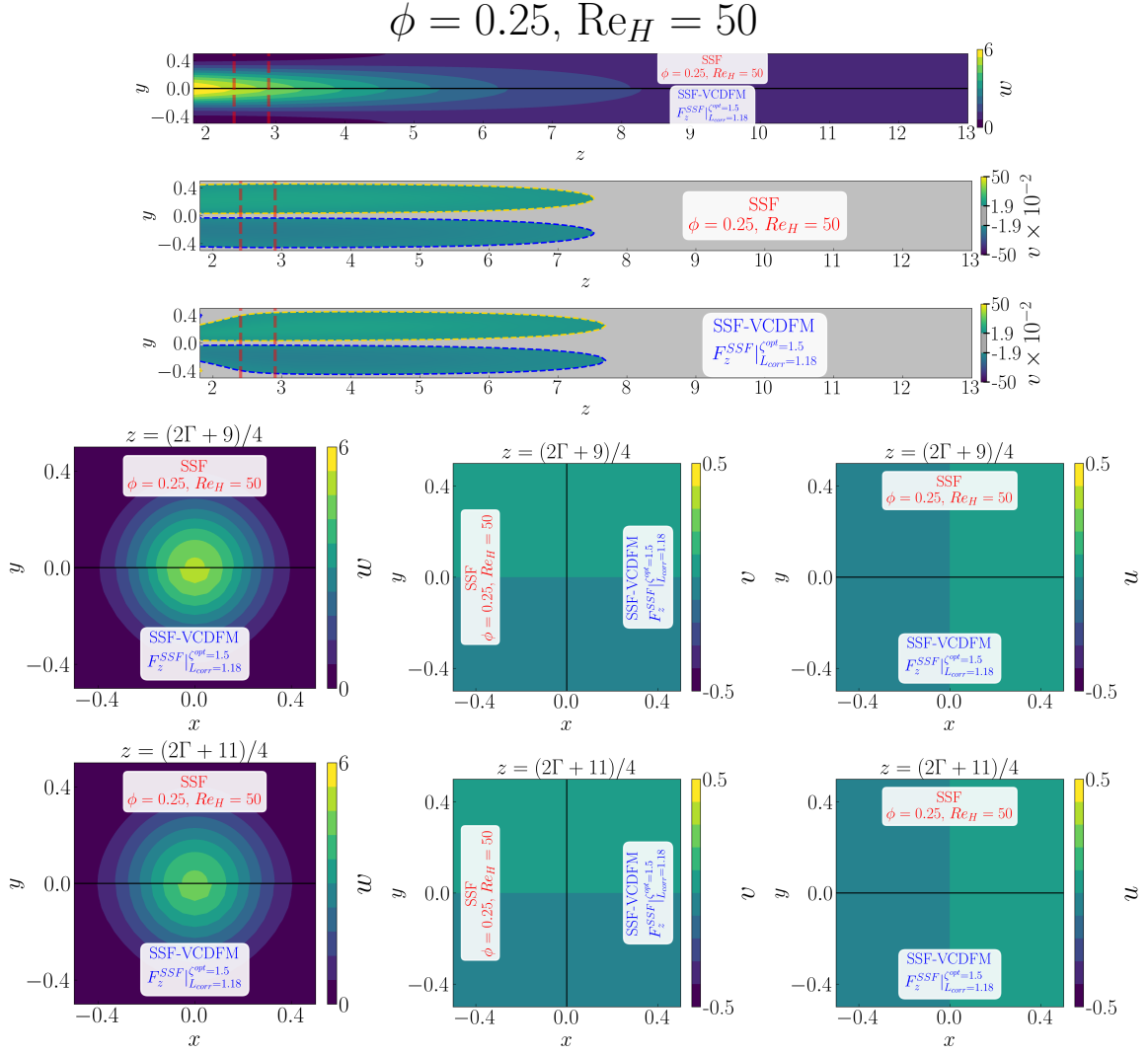


Figure D.210: Detailed comparison between the geometry-resolved small-scale flow (SSF) and SSF-VCDFM using F_z^{SSF} for the 1×1 SRG at $\phi = 0.25$ and $\text{Re}_H = 50$. The upper panels compare the streamwise velocity w and transverse velocity v in the central (z, y) -plane, restricted to the correction region. The lower panels compare the corresponding transverse (x, y) -plane distributions of w , v , and u , from left to right, at $z = (2\Gamma + 9)/4$ in the first row and $z = (2\Gamma + 11)/4$ in the second row.

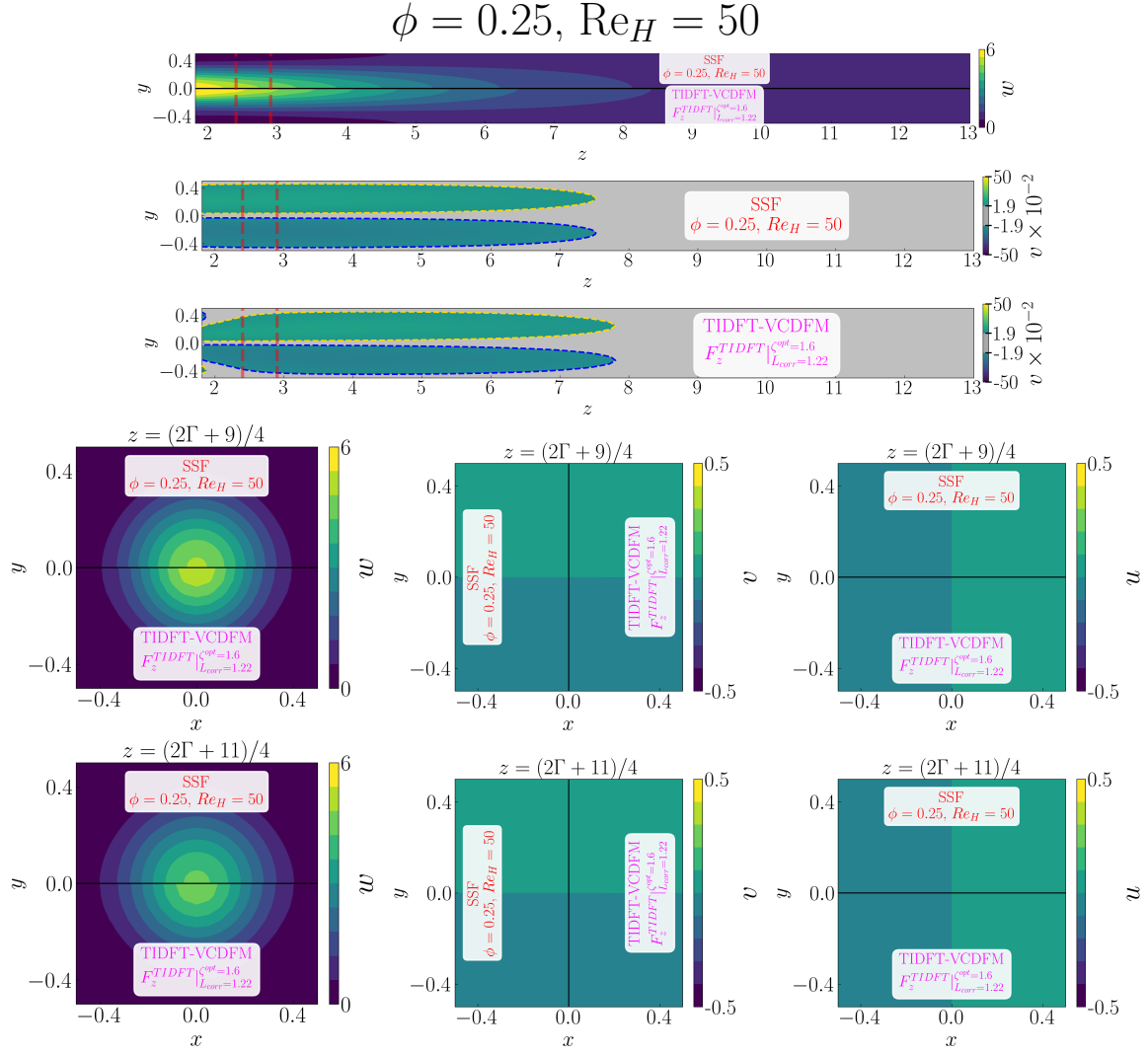


Figure D.211: Detailed comparison between the geometry-resolved small-scale flow (SSF) and TIDFT-VCDFM using F_z^{TIDFT} for the 1×1 SRG at $\phi = 0.25$ and $\text{Re}_H = 50$. The upper panels compare the streamwise velocity w and transverse velocity v in the central (z, y) -plane, restricted to the correction region. The lower panels compare the corresponding transverse (x, y) -plane distributions of w , v , and u , from left to right, at $z = (2\Gamma + 9)/4$ in the first row and $z = (2\Gamma + 11)/4$ in the second row.

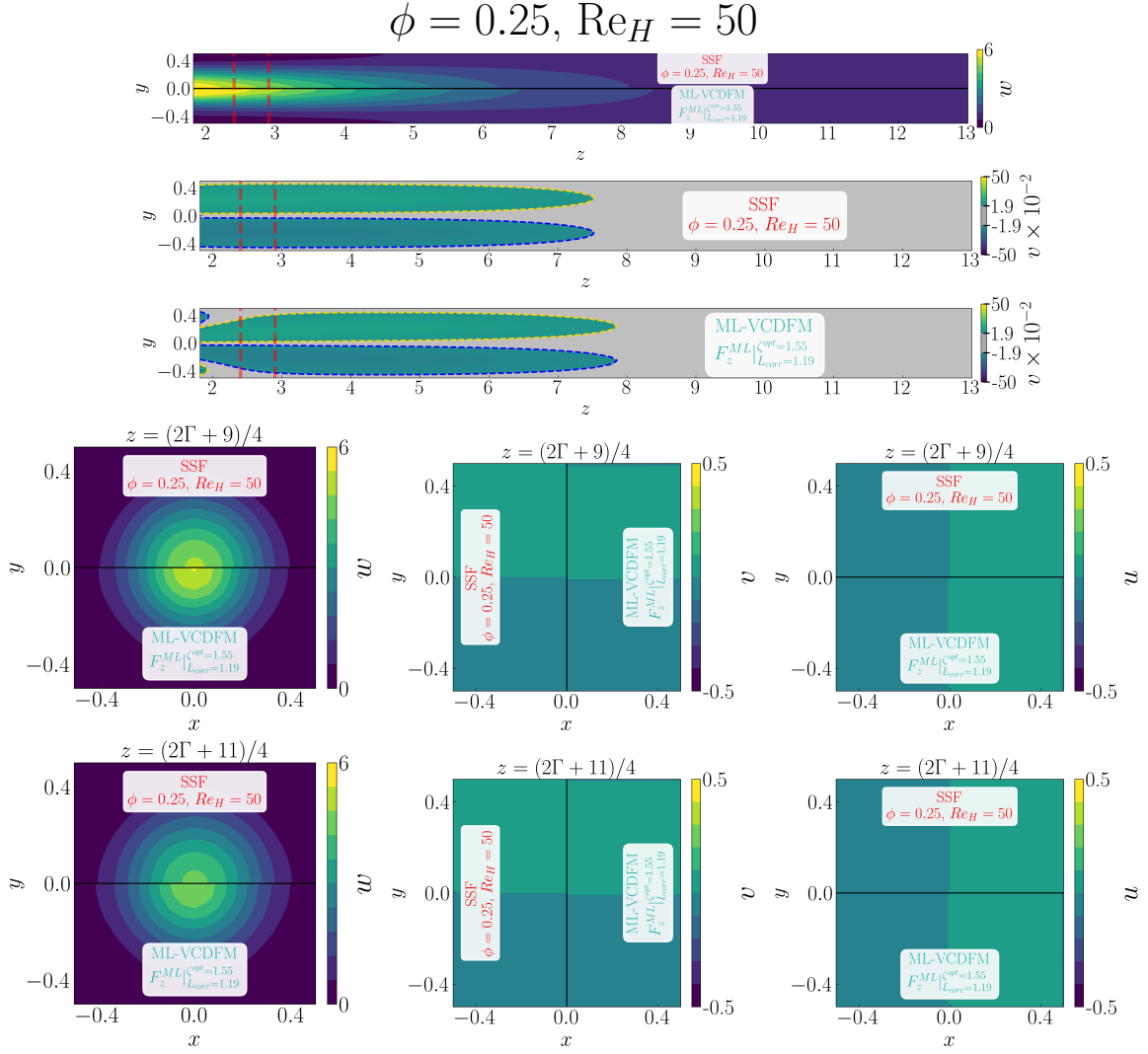


Figure D.212: Detailed comparison between the geometry-resolved small-scale flow (SSF) and ML-VCDFM using F_z^{ML} for the 1×1 SRG at $\phi = 0.25$ and $\text{Re}_H = 50$. The upper panels compare the streamwise velocity w and transverse velocity v in the central (z, y) -plane, restricted to the correction region. The lower panels compare the corresponding transverse (x, y) -plane distributions of w , v , and u , from left to right, at $z = (2\Gamma + 9)/4$ in the first row and $z = (2\Gamma + 11)/4$ in the second row.

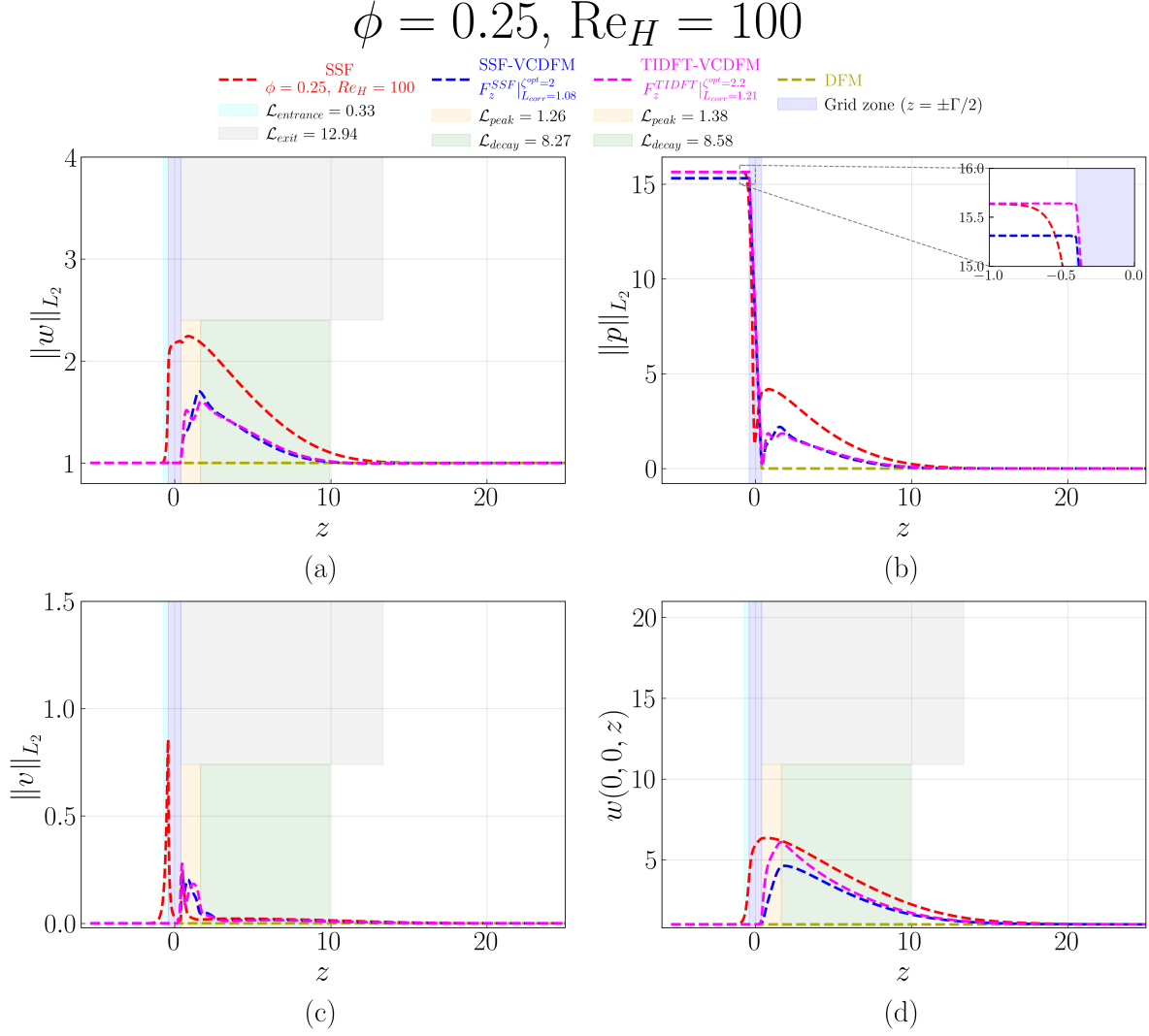


Figure D.213: Streamwise evolution of (a) $\|w\|_{L_2}$, (b) $\|p\|_{L_2}$, (c) $\|v\|_{L_2}$, and (d) centerline velocity $w(0,0,z)$ for the 1×1 SRG at $\phi = 0.25$ and $\text{Re}_H = 100$. The figure compares the SSF, DFM, SSF-VCDFM, and TIDFT-VCDFM. Shading marks the grid and the characteristic entrance, exit, peak, and decay regions. Only the SSF-based $\mathcal{L}_{\text{peak}}$ and $\mathcal{L}_{\text{decay}}$ are marked on the plot; the corresponding TIDFT-VCDFM values appear in the legend.

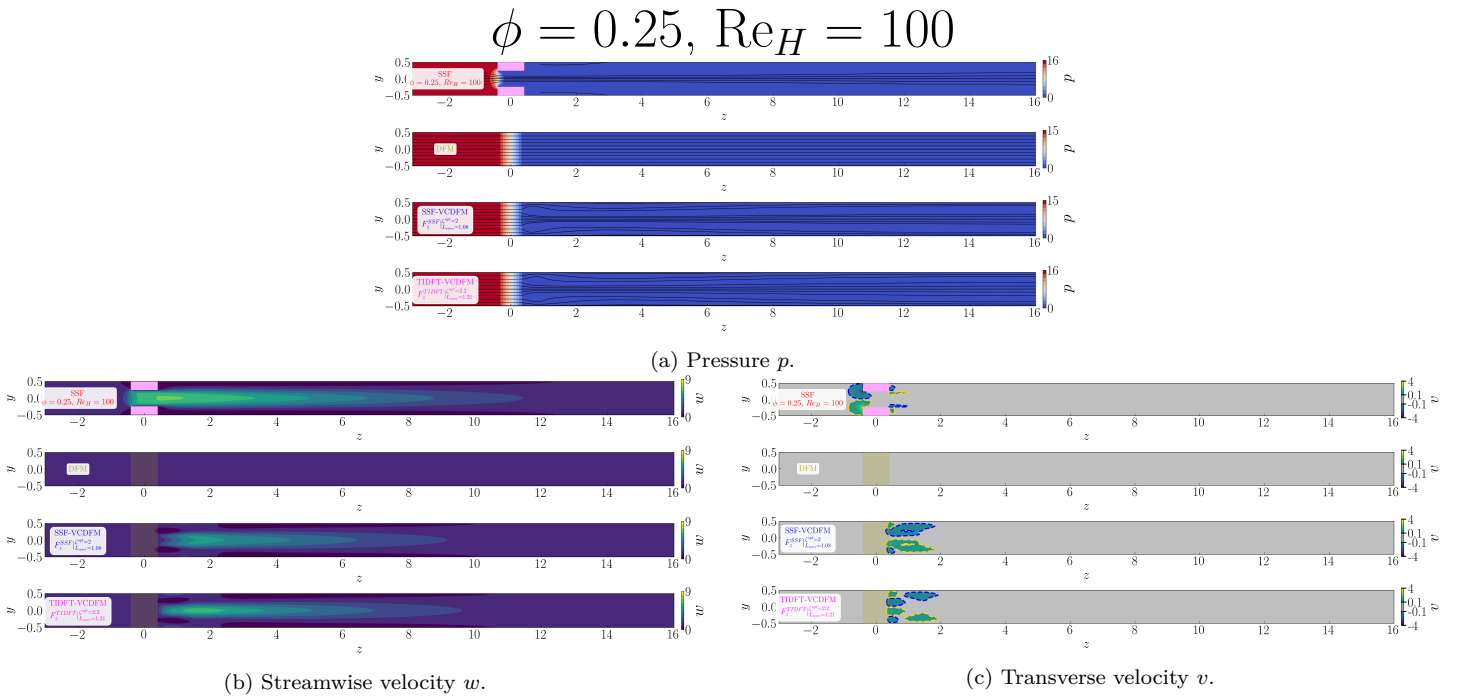


Figure D.214: Flow organization in the streamwise (z, y) -plane for the 1×1 SRG at $\phi = 0.25$, $\text{Re}_H = 100$, and $\alpha = 0^\circ$. The blocks show (a) pressure p with streamlines, (b) streamwise velocity w , and (c) transverse velocity v . Within each block, the rows show, from top to bottom, the SSF, DFM, SSF-VCDFM, and TIDFT-VCDFM. Pink shading marks the grid walls in the SSF; olive shading marks the equivalent porous zone in the model solutions.

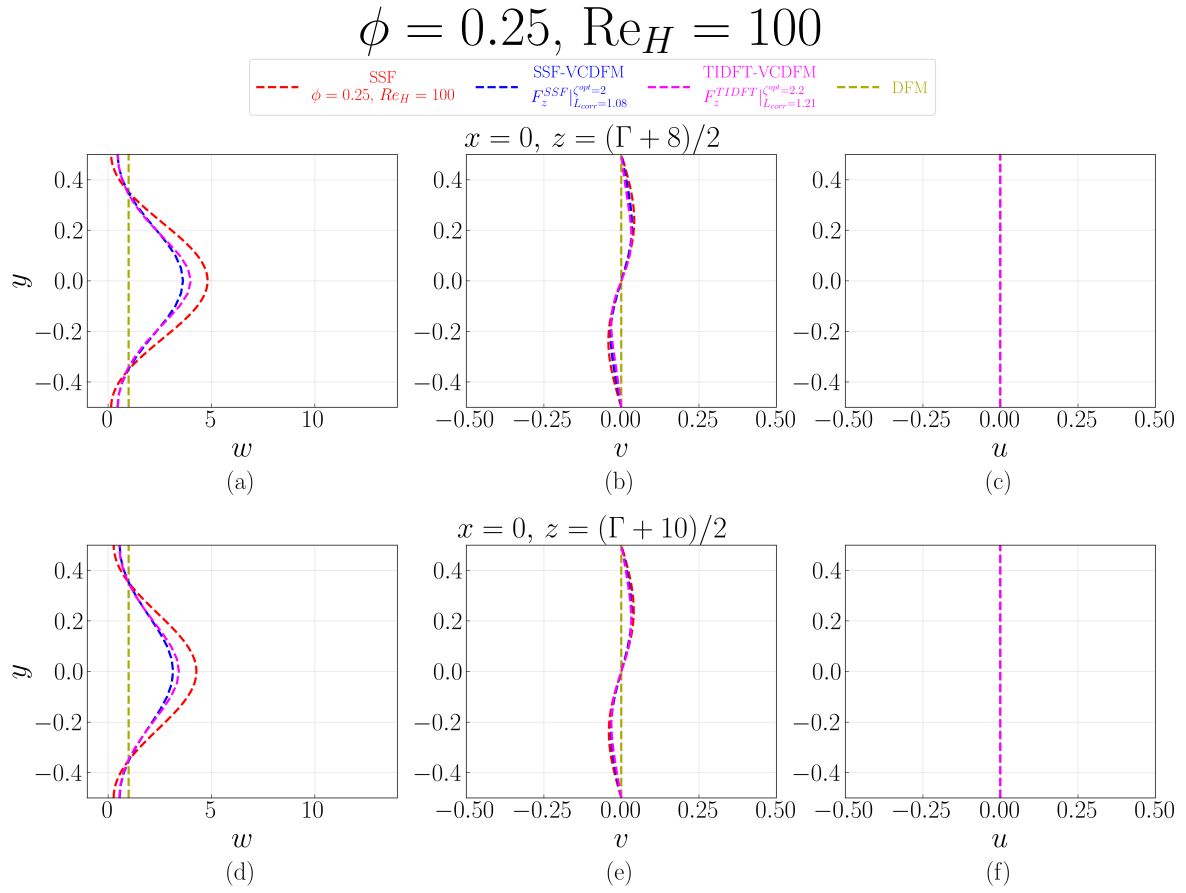


Figure D.215: Velocity profiles at $x = 0$ for the 1×1 SRG at $\phi = 0.25$ and $\text{Re}_H = 100$, comparing the SSF, DFM, SSF-VCDFM, and TIDFT-VCDFM. Profiles of $w(y)$, $v(y)$, and $u(y)$ are shown at $z = (\Gamma + 6)/2$ in (a–c) and $z = (\Gamma + 8)/2$ in (d–f).

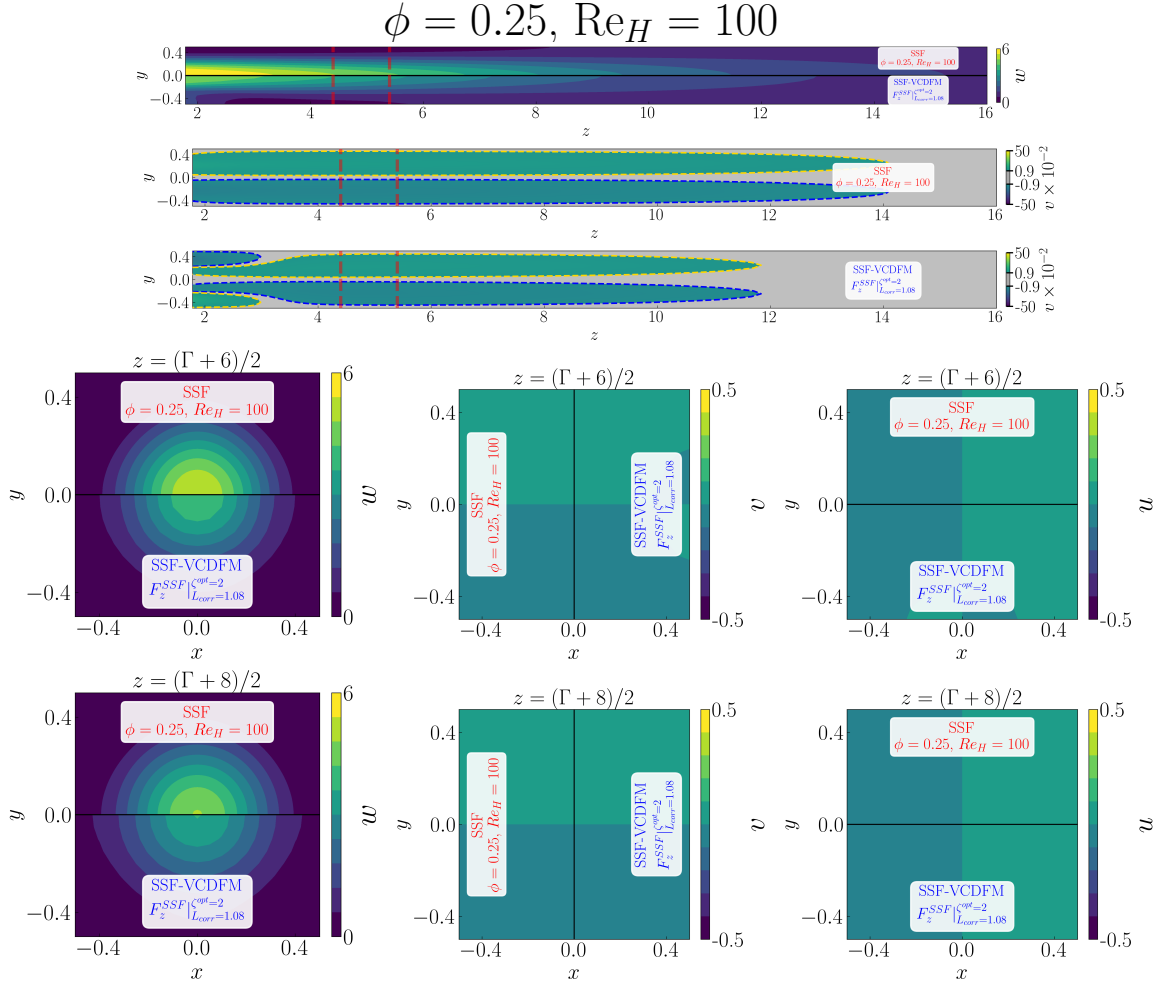


Figure D.216: Detailed comparison between the geometry-resolved small-scale flow (SSF) and SSF-VCDFM using F_z^{SSF} for the 1×1 SRG at $\phi = 0.25$ and $\text{Re}_H = 100$. The upper panels show the downstream flow in the streamwise (z, y) -plane, restricted to the correction region. The lower panels show the corresponding transverse (x, y) -plane distributions at $z = (\Gamma + 6)/2$ and $z = (\Gamma + 8)/2$.

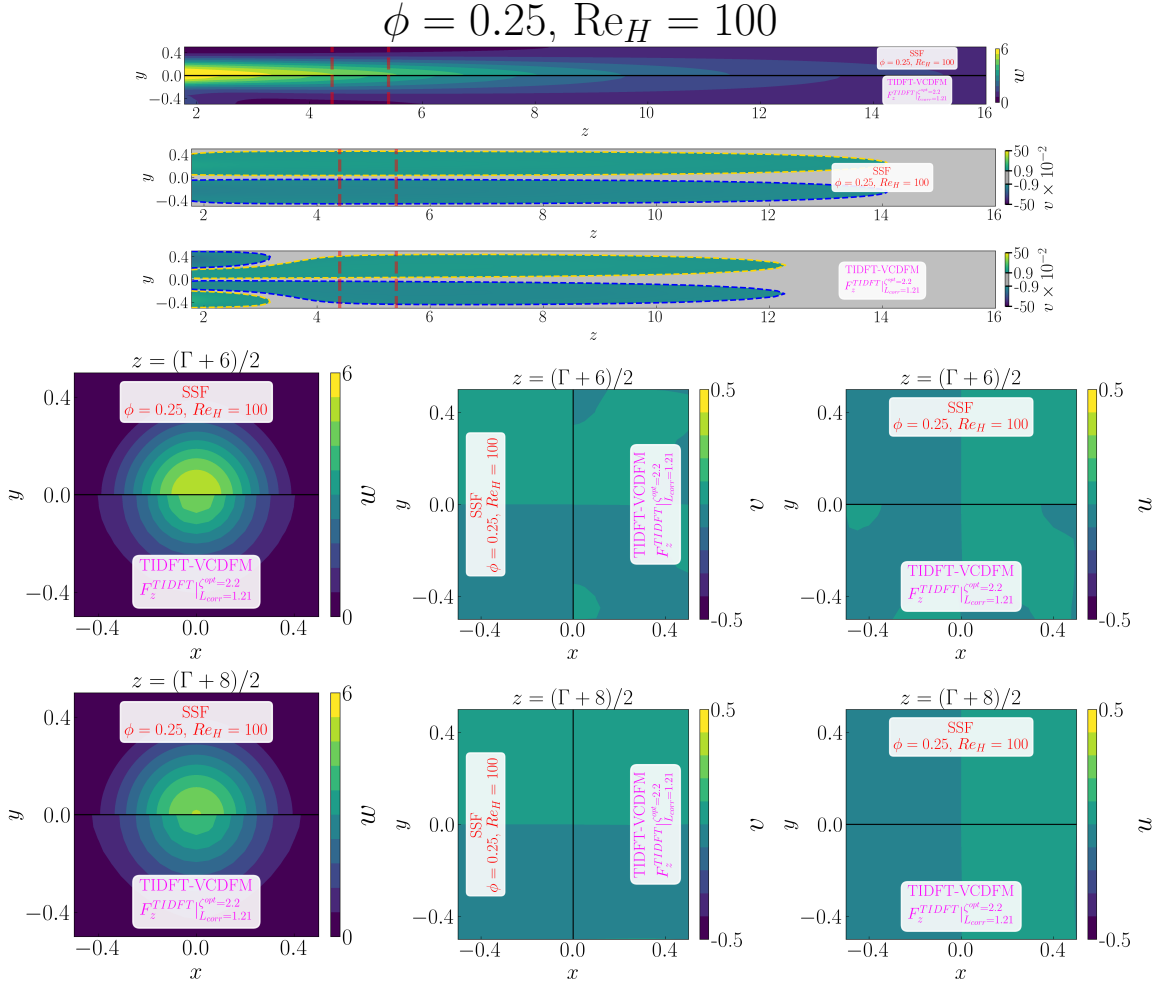


Figure D.217: Detailed comparison between the geometry-resolved small-scale flow (SSF) and TIDFT-VCDFM using F_z^{TIDFT} for the 1×1 SRG at $\phi = 0.25$ and $\text{Re}_H = 100$. The upper panels compare the downstream flow in the central (z, y) -plane, restricted to the correction region. The lower panels compare the corresponding transverse (x, y) -plane distributions at $z = (\Gamma + 6)/2$ and $z = (\Gamma + 8)/2$.

D.3 Detailed Validation Cases in the Breakdown Regime

This section provides detailed comparisons between the geometry-resolved small-scale flow (SSF) and the SSF-VCDFM, TIDFT-VCDFM, and, where available, ML-VCDFM for configurations identified outside the effective validity regime. These cases are included to characterize the onset of model breakdown and identify the flow features responsible for the loss of predictive accuracy of the VCDFM. For each case, the comparison includes cross-sectional L_2 -norms, velocity profiles, downstream organization in the (y, z) -plane, and transverse (x, y) -plane distributions.

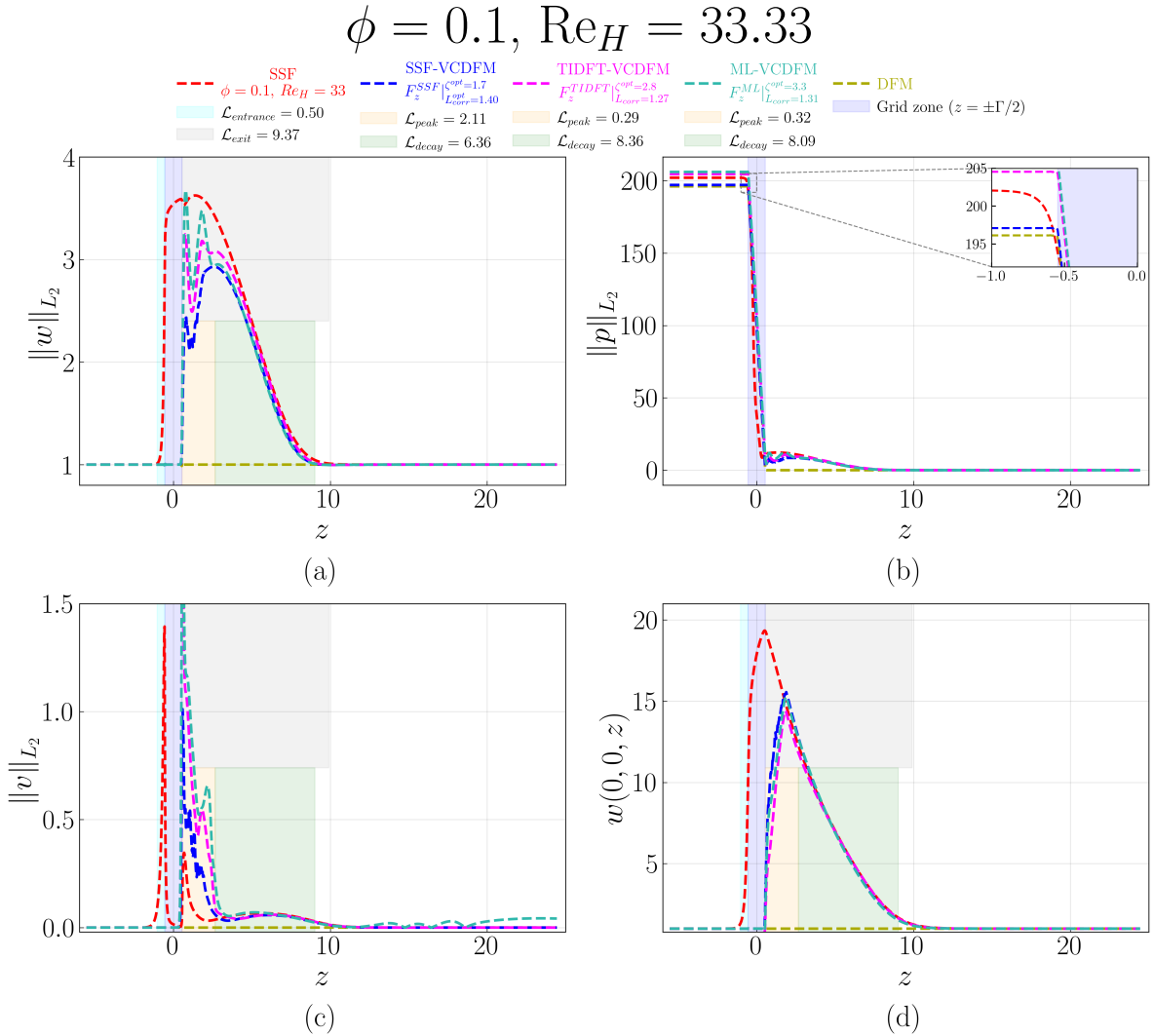


Figure D.218: Streamwise evolution of (a) $\|w\|_{L_2}$, (b) $\|p\|_{L_2}$, (c) $\|v\|_{L_2}$, and (d) centerline velocity $w(0,0,z)$ for the 1×1 SRG at $\phi = 0.1$ and $\text{Re}_H = 33.33$. The figure compares the SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM. Shading marks the grid and the characteristic entrance, exit, peak, and decay regions. Only the SSF-based $\mathcal{L}_{\text{peak}}$ and $\mathcal{L}_{\text{decay}}$ are marked on the plot; the corresponding TIDFT-VCDFM and ML-VCDFM values appear in the legend.

$$\phi = 0.1, \text{Re}_H = 33.33$$

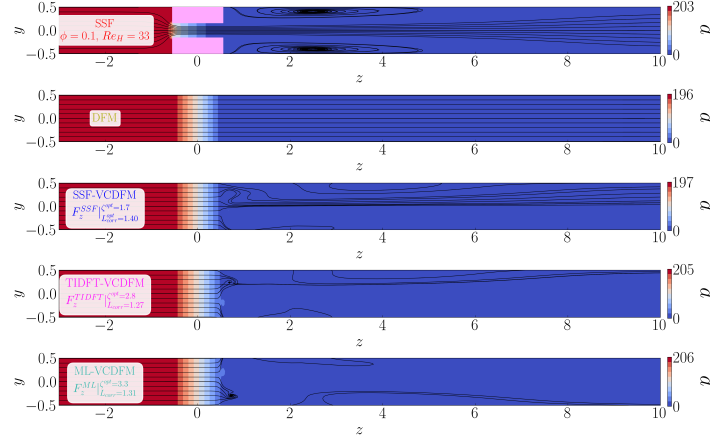
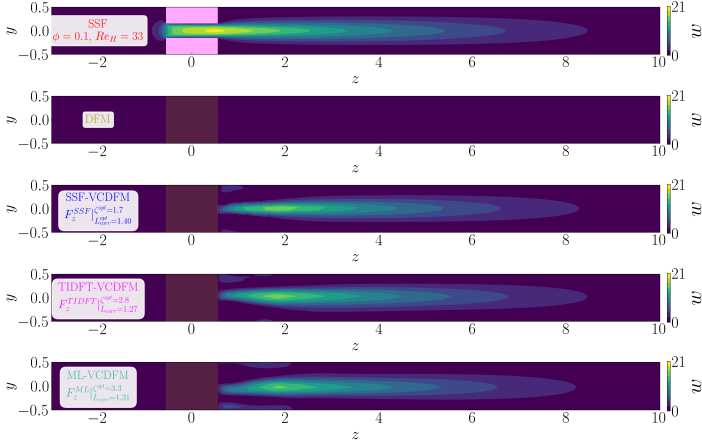
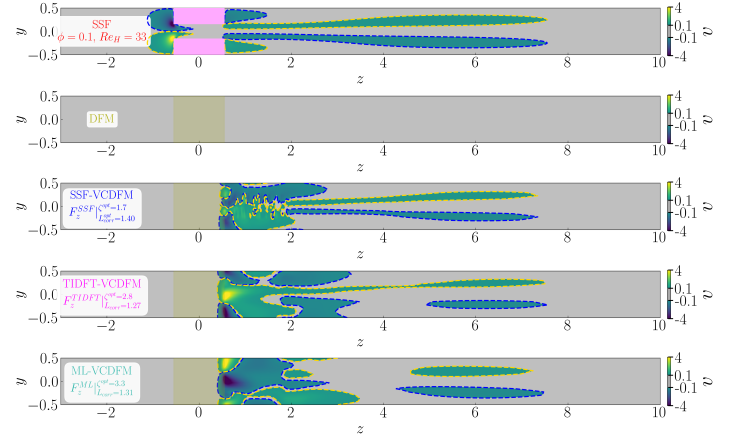

 (a) Pressure p .

 (b) Streamwise velocity w .

 (c) Transverse velocity v .

Figure D.219: Flow organization in the streamwise (z, y) -plane for the 1×1 SRG at $\phi = 0.1$, $\text{Re}_H = 33.33$, and $\alpha = 0^\circ$. The blocks show (a) pressure p with streamlines, (b) streamwise velocity w , and (c) transverse velocity v . Within each block, the rows show, from top to bottom, the SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM. Pink shading marks the grid walls in the SSF; olive shading marks the equivalent porous zone in the model solutions.

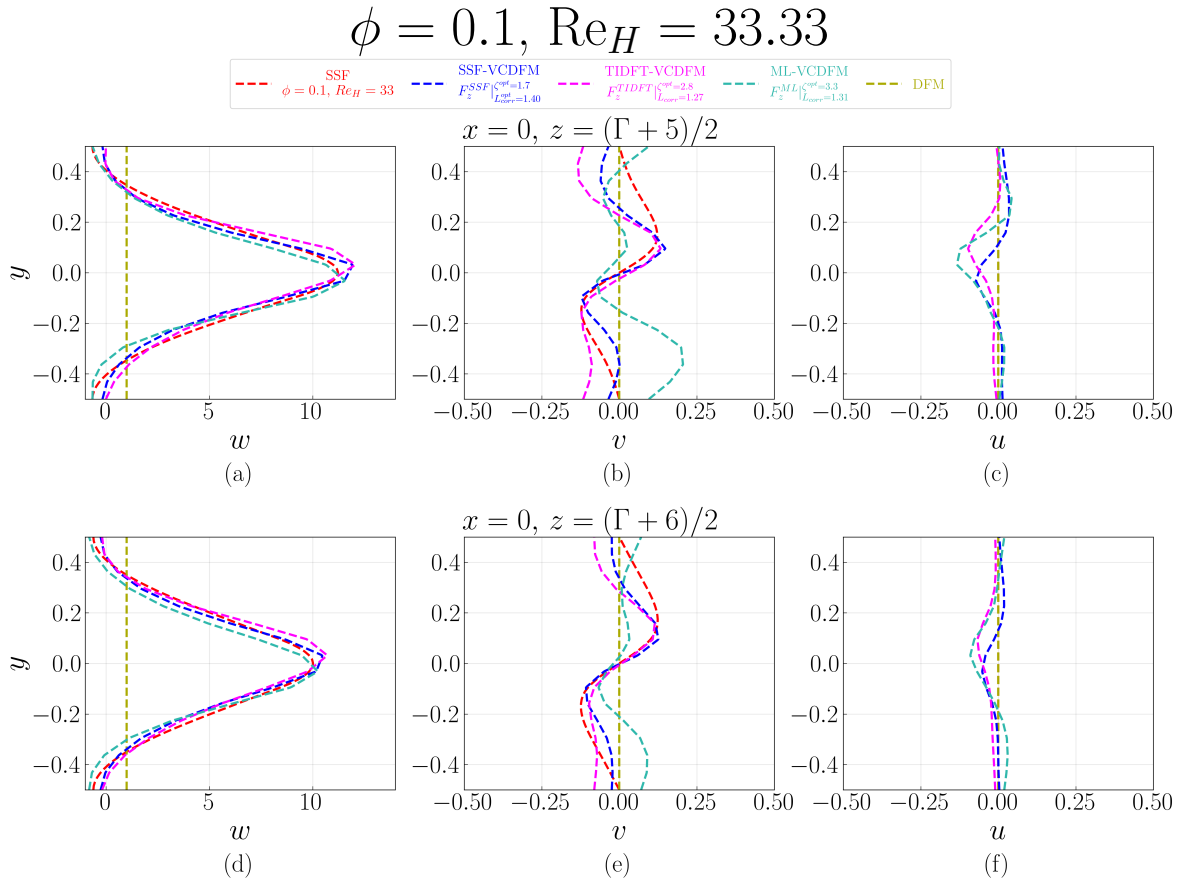


Figure D.220: Velocity profiles at $x = 0$ for the 1×1 SRG at $\phi = 0.1$ and $\text{Re}_H = 33.33$, comparing the SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM. Profiles of $w(y)$, $v(y)$, and $u(y)$ are shown at $z = (\Gamma + 5)/2$ in (a–c) and $z = (\Gamma + 6)/2$ in (d–f).

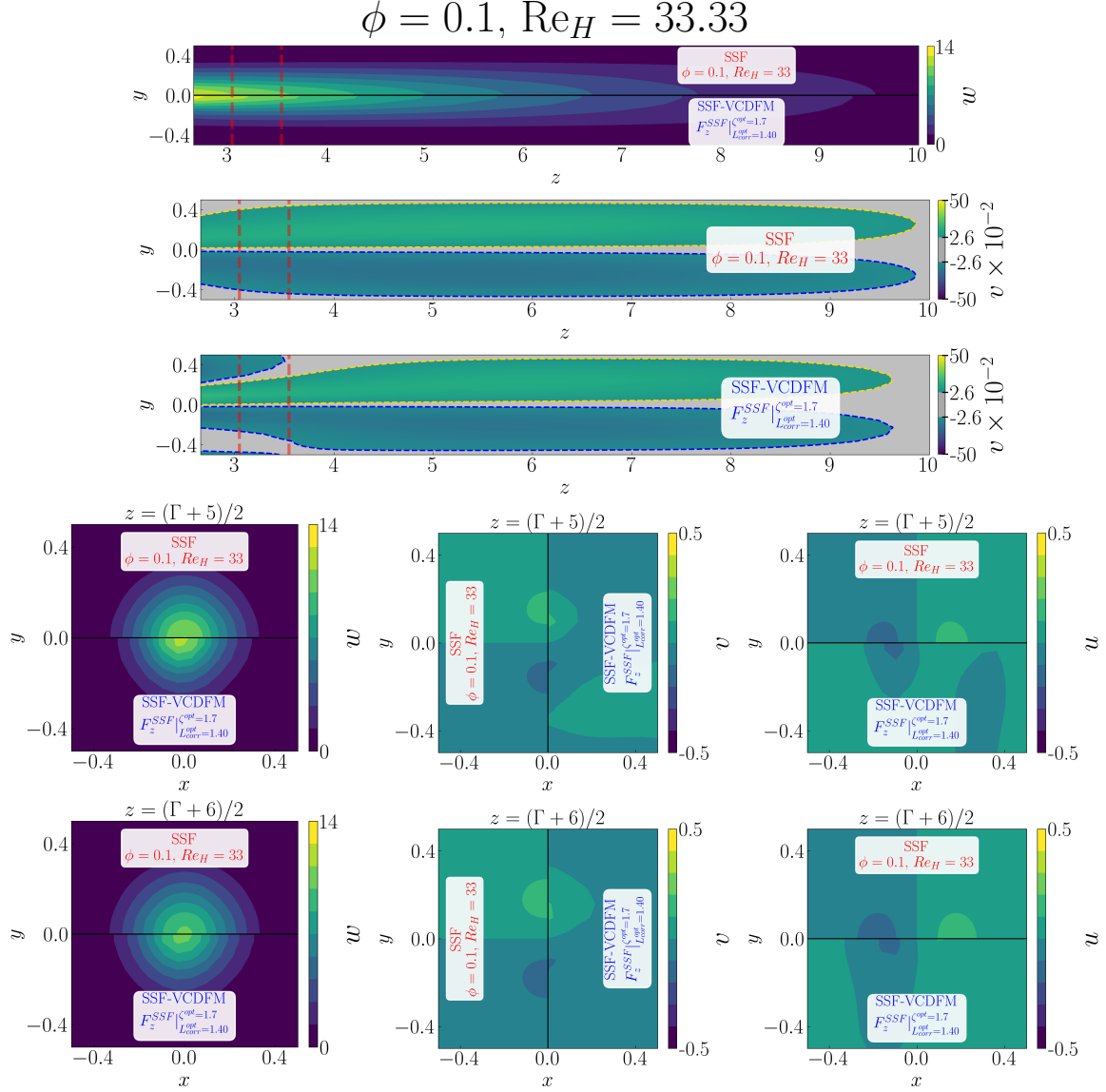


Figure D.221: Detailed comparison between the geometry-resolved small-scale flow (SSF) and SSF-VCDFM using F_z^{SSF} for the 1×1 SRG at $\phi = 0.1$ and $Re_H = 33.33$. The upper panels compare the streamwise velocity w and transverse velocity v in the central (z, y) -plane, restricted to the correction region. The lower panels compare the corresponding transverse (x, y) -plane distributions of w , v , and u , from left to right, at $z = (\Gamma + 5)/2$ in the first row and $z = (\Gamma + 6)/2$ in the second row.

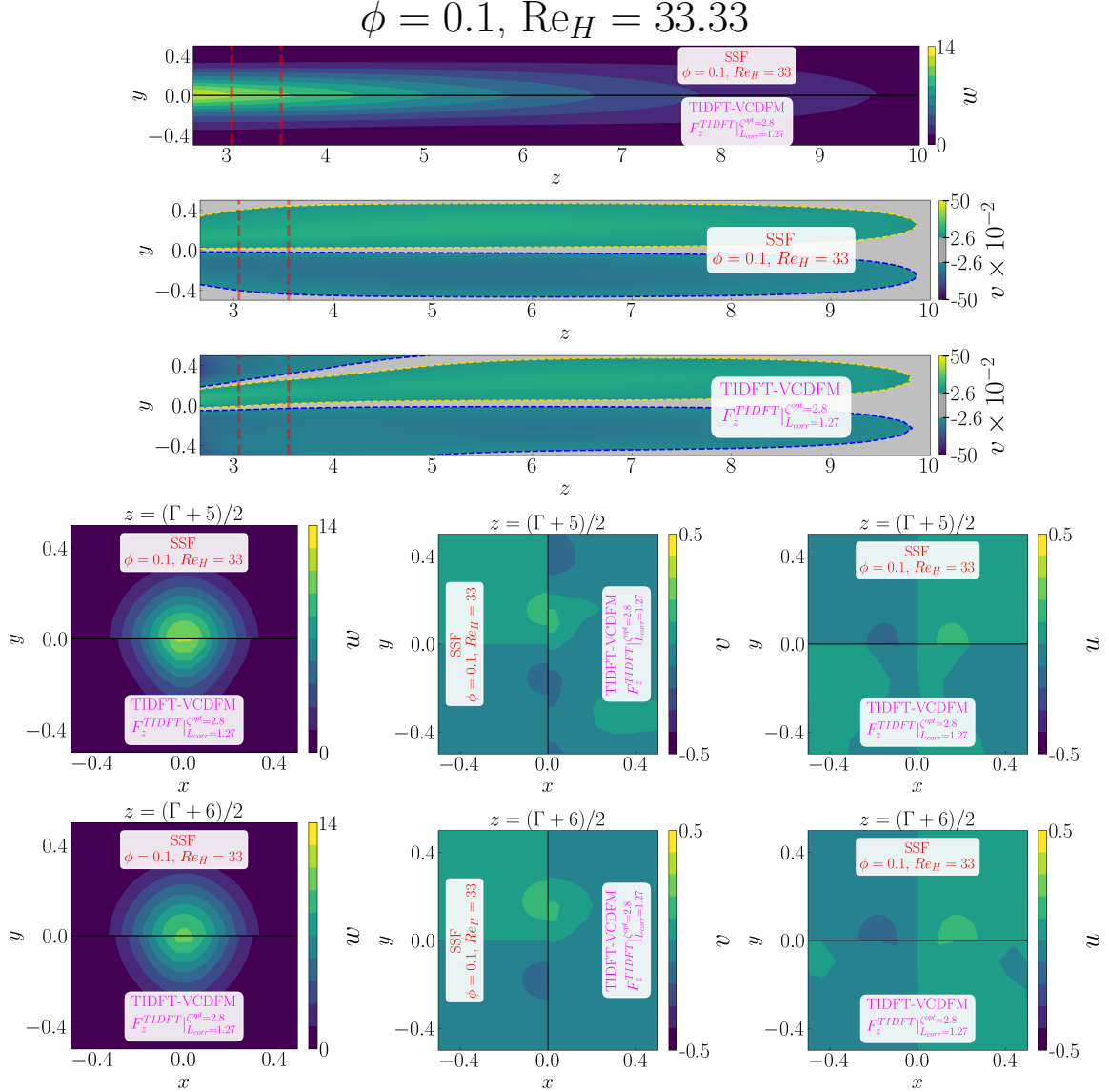


Figure D.222: Detailed comparison between the geometry-resolved small-scale flow (SSF) and TIDFT-VCDFM using F_z^{TIDFT} for the 1×1 SRG at $\phi = 0.1$ and $\text{Re}_H = 33.33$. The upper panels compare the streamwise velocity w and transverse velocity v in the central (z, y) -plane, restricted to the correction region. The lower panels compare the corresponding transverse (x, y) -plane distributions of w , v , and u , from left to right, at $z = (\Gamma + 5)/2$ in the first row and $z = (\Gamma + 6)/2$ in the second row.